Contact-Scheduled Kinematic Reference Synthesis for Microspine Wall Climbing without Motion Capture

Method section draft

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1 Problem statement

1.1 Setting and notation

We consider a humanoid (PHYBot C2) climbing a planar wall inclined at angle α from the floor ($\alpha = 90°$: vertical). The robot has $n_j = 21$ actuated joints $\boldsymbol{\theta} \in \mathbb{R}^{n_j}$ (6 per leg, 4 per wristless arm, 1 waist yaw) and a floating base $(\boldsymbol{p}_b, R_b) \in \mathrm{SE}(3)$. Four microspine pads $h \in \mathcal{H} = \{\mathrm{LH}, \mathrm{RH}, \mathrm{LF}, \mathrm{RF}\}$ are mounted on the hands and shins. Each pad sits on a passive three-hinge gimbal with angles $\boldsymbol{\gamma}_h \in \mathbb{R}^3$, mechanical stops $|\gamma_{h,k}| \le \gamma_{\mathrm{stop}} = 3°$, and a weak return spring ($0.7\,\mathrm{N\,m/rad}$). The feet additionally carry a rigid toe tip that may press, but not adhere to, the wall.

Positions are expressed in the canonical wall frame: the wall surface is the plane $x = x_w$, $\hat{\boldsymbol{z}}$ points up the wall, and $\hat{\boldsymbol{n}} = -\hat{\boldsymbol{x}}$ is the outward wall normal (pointing at the robot). A tilted wall is obtained by rotating this frame about the wall base line; gravity stays physically vertical. On an engaged pad the *engage frame* $(\hat{\boldsymbol{t}}, \hat{\boldsymbol{l}}, \hat{\boldsymbol{n}})$ has $\hat{\boldsymbol{t}}$ along the rake direction (down-wall), $\hat{\boldsymbol{l}} = \hat{\boldsymbol{n}} \times \hat{\boldsymbol{t}}$ lateral.

A microspine pad holds a load $\boldsymbol{F} = (f_t, f_l, f_n)$ only inside the closed, direction-dependent limit surface of Wang, Jiang and Cutkosky (IJRR 2017): capacity peaks along the rake shear $f_t > 0$, tension $f_n > 0$ is admissible only alongside shear, and against-rake and pure normal pull-off hold almost nothing. A pad seats only after a small down-wall stroke under compression (the *rake*) and recruits its full capacity over $\tau_{\mathrm{ramp}} \approx 0.25\,\mathrm{s}$ of loaded dwell.

A reference motion is a tuple

$$\mathcal{M} = \big(\boldsymbol{q}(t),\ \boldsymbol{\sigma}(t),\ \{A_h(t)\}_{h\in\mathcal{H}},\ \boldsymbol{\beta}(t)\big), \qquad t \in [0, T], \tag{1}$$

with configuration $\boldsymbol{q} = (\boldsymbol{p}_b, R_b, \boldsymbol{\theta})$, contact schedule $\boldsymbol{\sigma}(t) \in \{0,1\}^4$ ($\sigma_h = 1$: pad h latched), anchor poses $A_h = (\boldsymbol{a}_h, R_h^a) \in \mathrm{SE}(3)$ (the weld target while latched, the landing target while swinging), and a toe-brace schedule $\boldsymbol{\beta}(t) \in \{0,1\}^2$. The reference is consumed by reinforcement learning as an imitation target for *shape and schedule only*: no reward tracks the base position in the world, and force allocation and balance are left to the learned policy.

1.2 Why motion capture does not transfer

Imitation pipelines for humanoids obtain $\boldsymbol{q}(t)$ from human motion capture followed by kinematic retargeting. Two obstacles make this route unavailable here.

Data. Adhesive wall climbing with palm and shin pads is not a human activity; no capture corpus exists, and staged approximations (bouldering, ladder climbing) exercise fingers, wrists and toes the robot does not have and load the holds through friction geometry the pads cannot reproduce.

Embodiment optimality. Even with data, the human optimum is not the robot optimum. The feasible set of the robot is dictated by (i) a 4-DoF wristless arm that cannot satisfy an arbitrary 6-DoF hand pose, (ii) a pad that must stay flat on the wall and may rotate only within the 3° gimbal travel once latched, (iii) an adhesion envelope that is essentially one-sided: the pad must be loaded along its rake direction and must never be dragged across the surface while unloaded, and (iv) a recruitment time that forbids fast regrips. Retargeting projects a human trajectory onto the robot's kinematic chain, but not onto this *contact-feasibility* set; the projected motion drags latched pads, loads them against the rake, or transfers weight onto immature anchors. In our experience, imitating references that lack an explicit schedule and the physics gates of Sec. 5 produced degenerate gaits: policies welded two limbs permanently and dragged the others (Sec. 7).

1.3 Requirements

A usable reference must satisfy:

(R1) **Explicit schedule.** $\boldsymbol{\sigma}$ and $\{A_h\}$ are part of the reference, so that the policy can be rewarded for latching the right pad at the right place and time.

(R2) **Kinematic closure.** While $\sigma_h = 1$ the pad is stationary at A_h (a latched pad physically cannot move), the limb may rotate relative to it only within the gimbal travel, all joints stay strictly inside their limits, and released pads and toes do not penetrate the wall.

(R3) **Static feasibility.** At every frame there exists a force allocation on the scheduled contacts that balances gravity inside the limit surface and the actuator torque limits.

(R4) **Timing feasibility.** Dwell times respect the engage pipeline and the recruitment ramp.

(R5) **Periodicity.** One cycle ends exactly at its start translated by the step height h up the wall, so the clip loops.

(R6) **Determinism and low cost.** Regeneration takes minutes and is independent of runtime physics parameters, so that the reference can sit inside a design loop.

2 Method overview

The reference is synthesized in three stages (Fig. [TBD]). *Choreography* (Sec. 3) fixes the contact schedule and a small set of task-space intents from the robot's own stance. *Lexicographic constrained inverse kinematics* (Sec. 4) turns the intents into joint trajectories frame by frame, with the latched contacts and the mechanical limits as hard constraints and every intent in their null space. *Physics gates* (Sec. 5) certify static and timing feasibility and close the loop on the choreography parameters. Section 6 states how the policy consumes the result.

3 Choreography: contact schedule and task-space intents

Stance and anchors. The cycle starts from the climb stance $\boldsymbol{q}^0$ (all four pads latched, arms overhead, legs splayed), obtained once by an inverse-kinematics stance search that pins the four pad frames flat on the wall plane at a chosen hand height and base standoff and minimizes posture cost. All anchors are derived from it by forward kinematics,

$$A_h^0 = \mathrm{FK}_h(\boldsymbol{q}^0), \qquad A_h^+ = \left(\boldsymbol{a}_h^0 + h\,\hat{\boldsymbol{z}},\ R_h^{a,0}\right), \qquad h = 0.03\,\mathrm{m}, \tag{2}$$

so that no world coordinate is authored by hand except the wall plane.

Schedule. One cycle is a four-beat crawl,

$$\text{hold} \to \text{LF swing} \to \text{hold} \to \text{RF swing} \to \text{hold} \to \text{LH swing} \to \text{hold} \to \text{RH swing} \to \text{hold},$$

subject to the rules that encode the pad physics:

(S1) at most one pad is released at any time, so $\sum_h \sigma_h \geq 3$;

(S2) the support set always contains at least one hand and one foot, preserving the upper-tension/lower-compression couple that balances the wall-normal gravity moment;

(S3) every hand swing starts with a short all-four load-transfer dwell, and every dwell before a release lasts at least τ_{ramp};

(S4) during a foot swing the toe of the released foot braces the wall as a compression contact ($\beta_h = 1$), so the unloaded spine pad never slides;

(S5) every landing ends with a rake: the pad first touches the wall d_{rake} above its anchor and is dragged straight down the surface onto it before σ_h flips to 1.

Segment durations N_k (frames at $f_s = 50\,\text{Hz}$) are choreography parameters; Table 1 lists the shipped cycle.

Segment	Frames	Duration [s]	Released pad / brace
hold_start	15	0.30	none (left toe plants)
left_foot_swing	38	0.76	LF, left toe braces
foot_transfer_hold	15	0.30	none (retract left toe, plant right)
right_foot_swing	38	0.76	RF, right toe braces
seat_hold	15	0.30	none (retract right toe)
left_hand_swing	45	0.90	LH after 0.1 s transfer dwell
regrip_hold	15	0.30	none
right_hand_swing	45	0.90	RH after 0.1 s transfer dwell
cycle	226	4.52	$h = 30\,\text{mm}$, $6.6\,\text{mm/s}$

Table 1: Contact schedule of the shipped reference cycle (`climb_cycle_moment_safe_paced`).

Time parameterization. Within a segment of N frames the progress variable is

$$s(t) = S\Big(\tfrac{t}{N-1}\Big), \qquad S(x) = 6x^5 - 15x^4 + 10x^3, \tag{3}$$

which has zero velocity and acceleration at both ends.

Swing path. For the released pad with start pose $(\boldsymbol{p}^0, R^0) = \text{FK}_h(\boldsymbol{q})$ at release and target anchor $(\boldsymbol{p}^+, R^a)$, the last N_r frames are the rake and the first $N_a = N - N_r$ the approach arc,

$$\boldsymbol{c} = \tfrac{1}{2}(\boldsymbol{p}^0 + \boldsymbol{p}^+) + b\,\hat{\boldsymbol{n}} + \ell\,\hat{\boldsymbol{y}}_{\text{side}}, \qquad \boldsymbol{r} = d_{\text{rake}}\,\hat{\boldsymbol{z}}, \tag{4}$$

$$\boldsymbol{p}^*(t) = \begin{cases} (1-s_a)^2\,\boldsymbol{p}^0 + 2(1-s_a)s_a\,\boldsymbol{c} + s_a^2\,(\boldsymbol{p}^+ + \boldsymbol{r}), & t < N_a, \quad s_a = S\big(\min(\tfrac{t}{N_a-1}, 1)\big), \\ \boldsymbol{p}^+ + \big(1 - S(\tfrac{t-N_a+1}{N_r})\big)\,\boldsymbol{r}, & t \geq N_a, \end{cases} \tag{5}$$

with off-wall bulge b (hands 0.04 m, feet 0.03 m), lateral offset ℓ (default 0) and $d_{\text{rake}} = 2\,\text{mm}$. The orientation target of a foot is the geodesic interpolation $R^*(s) = \exp\big(s\log(R^a R^{0\top})\big)R^0$; a hand holds R^0 and its orientation weight ramps toward landing (Sec. 4.3).

Toe brace. With toe tip position $\boldsymbol{p}_{\text{toe}}$ and radius r_{toe} the brace target is $x^*_{\text{toe}}(t) = (1-\pi(t))\, x_{\text{plant}} + \pi(t)\,(x_w - r_{\text{toe}})$, where the press profile $\pi(t)$ ramps $0 \to 1$ over 0.06 s, holds 1, and returns to 0 during the rake; y, z ride freely.

Base and posture intents. During a swing the base target and the posture reference blend from their values at the segment start,

$$\boldsymbol{p}_b^*(s) = (1-s)\,\boldsymbol{p}_b(t_0) + s\,\boldsymbol{p}_b^{\text{end}}, \qquad \boldsymbol{\theta}^*(s) = (1-s)\,\boldsymbol{\theta}(t_0) + s\,\boldsymbol{\theta}^0, \tag{6}$$

with $\boldsymbol{p}_b^{\text{end}} = \boldsymbol{p}_b^0 - \delta_{\text{press}}\hat{\boldsymbol{n}} + \delta_{\text{lat}}\hat{\boldsymbol{y}}_{\text{side}} + \frac{k+1}{2} h\,\hat{\boldsymbol{z}}$ for the k-th foot swing ($\delta_{\text{press}} = 0.03$ m toward the wall, $\delta_{\text{lat}} = 0.025$ m toward the supporting side). These are intents, not constraints: the solver realizes them only as far as the latched contacts allow.

4 Lexicographic constrained inverse kinematics

Each frame is solved by iterating a two-level quadratic program in the tangent increment of the configuration until convergence. The design principle is that *physical facts are hard and intents are soft*: a latched pad does not move, a hinge does not pass its stop, a joint does not leave its range; everything the choreographer merely *wants* is realized in the null space of those facts.

4.1 Decision variables and hard bounds (level 0)

The step is

$$\delta\boldsymbol{q} = (\delta\boldsymbol{p}_b,\ \delta\boldsymbol{\phi}_b,\ \delta\boldsymbol{\theta},\ \delta\boldsymbol{\gamma}) \in \mathbb{R}^n, \qquad n = 6 + n_j + 3|\mathcal{H}| = 39, \tag{7}$$

where $\delta\boldsymbol{\phi}_b$ is the base rotation vector and $\delta\boldsymbol{\gamma}$ stacks the gimbal angles of all four pads. Bounds encode the mechanical limits and a trust region:

$$\boldsymbol{\theta}_{\text{lo}} + \boldsymbol{m} \ \le\ \boldsymbol{\theta} + \delta\boldsymbol{\theta} \ \le\ \boldsymbol{\theta}_{\text{hi}} - \boldsymbol{m}, \qquad \boldsymbol{m} = \mu\,(\boldsymbol{\theta}_{\text{hi}} - \boldsymbol{\theta}_{\text{lo}}),\ \mu = 0.01, \tag{8}$$

$$-\gamma_{\max} \le \gamma_{h,k} + \delta\gamma_{h,k} \le \gamma_{\max} \qquad \text{if } \sigma_h = 1, \quad \gamma_{\max} < \gamma_{\text{stop}}, \tag{9}$$

$$\gamma_{h,k} + \delta\gamma_{h,k} = 0 \qquad \text{if } \sigma_h = 0, \tag{10}$$

$$\|\delta\boldsymbol{q}\|_\infty \le \Delta. \tag{11}$$

The joint margin μ matches the 0.98 soft band the downstream policy clamps its targets to, so the reference never asks the policy to fight its own clamp. A latched pad may use its gimbal travel up to $\gamma_{\max}$ (default 2°), leaving $\gamma_{\text{stop}} - \gamma_{\max}$ of travel for the policy's own tracking error. A released pad's gimbal is pinned at zero: in the plant it is passive and unloaded, and the solver must not use it to fake a landing orientation. All of (8)–(11) are box constraints $\boldsymbol{l} \le \delta\boldsymbol{q} \le \boldsymbol{u}$.

4.2 Hard task-space constraints (level 1)

For every latched pad h ($\sigma_h = 1$) the pad frame, i.e. the body *below* the gimbal, is welded to its anchor. With $(\boldsymbol{p}_h, R_h) = \text{FK}_h(\boldsymbol{q})$, site Jacobians $J_h^p, J_h^r \in \mathbb{R}^{3\times n}$ (including the gimbal columns), and the world-frame small-angle error

$$\boldsymbol{e}_h^p = \boldsymbol{a}_h - \boldsymbol{p}_h, \qquad \boldsymbol{e}_h^r = \tfrac{1}{2}\sum_{k=1}^{3} \boldsymbol{r}_{h,k} \times \boldsymbol{r}^a_{h,k} \quad (\boldsymbol{r}_{h,k}, \boldsymbol{r}^a_{h,k}\text{: columns of } R_h, R^a_h), \tag{12}$$

the weld rows are

$$J_1\,\delta\boldsymbol{q} = \boldsymbol{e}_1, \qquad J_1 = \begin{bmatrix} J^p_{h_1} \\ J^r_{h_1} \\ \vdots \end{bmatrix}, \quad \boldsymbol{e}_1 = \begin{bmatrix} \boldsymbol{e}^p_{h_1} \\ \boldsymbol{e}^r_{h_1} \\ \vdots \end{bmatrix} \in \mathbb{R}^{6n_c}, \tag{13}$$

$n_c = \sum_h \sigma_h \in \{3, 4\}$. Wall clearance of released pads and toes is a linear inequality on the same level: for every contact sphere or toe tip c of a released pad with center $\boldsymbol{p}_c$, radius r_c and Jacobian J_c,

$$\hat{\boldsymbol{x}}^\top \big(\boldsymbol{p}_c + J_c\, \delta\boldsymbol{q}\big) \le x_w - r_c \quad \Longleftrightarrow \quad G\, \delta\boldsymbol{q} \le \boldsymbol{g}. \tag{14}$$

4.3 Soft objectives in the null space (level 2)

The intents form a weighted least-squares cost,

$$\begin{aligned} f_2(\delta\boldsymbol{q}) = \tfrac{1}{2}\sum_{i\in\mathcal{S}} \big\| W_i (J_i \delta\boldsymbol{q} - \boldsymbol{e}_i) \big\|^2 + \tfrac{1}{2} \big\| W_\theta (\delta\boldsymbol{\theta} - (\boldsymbol{\theta}^* - \boldsymbol{\theta})) \big\|^2 \\ + \tfrac{1}{2} \big\| W_{\lim} (\delta\boldsymbol{\theta} - (\boldsymbol{\theta}_{\text{bnd}} - \boldsymbol{\theta})) \big\|^2 + \tfrac{1}{2} w_\gamma^2 \|\boldsymbol{\gamma} + \delta\boldsymbol{\gamma}\|^2 + \tfrac{1}{2}\lambda^2 \|\delta\boldsymbol{q}\|^2, \end{aligned} \tag{15}$$

where $\mathcal{S}$ collects the swing-pad pose task, the toe-brace task and the base task; $W_\theta = \operatorname{diag}(w_\theta)$ pulls toward the posture reference; $W_{\lim}$ is nonzero only for joints inside a margin band of width $\mu_s(\theta_{\text{hi}} - \theta_{\text{lo}})$, $\mu_s = 0.10$, next to either stop and pulls them back to the band boundary θ_{bnd}; w_γ prefers the arm joints over the gimbal; and $\lambda^2 = 0.5$ is the Levenberg damping. Table 2 lists the weights. The swing hand's weights fade in over the first 30 % of the arc and grow again toward landing, $w = w_{\min} + (w_{\max} - w_{\min}) \max\big(1 - \min(s_a/0.3, 1),\, s_a^2\big)$, which spreads the re-equilibration caused by releasing a weld over the whole swing instead of a one-frame pop.

Task	Rows	w_{pos}	w_{rot}
latched pad (weld)	level 1, hard	–	–
swing foot pad	6	1000	100
swing hand pad	6	$500 \to 2000$	$50 \to 500$
toe brace	3 (position only)	(300, 30, 30)	–
base intent	6	30–100	2–50
posture reference w_θ	n_j	0.3–0.6 (RH swing 10)	–
limit-band spring $w_{\lim}$	active joints	3	–
gimbal return w_γ	12	3	–
damping λ	n	$\sqrt{0.5}$	–

Table 2: Task hierarchy and level-2 weights. Latched pads are hard constraints and carry no weight.

4.4 Two-stage solution, convergence and feasibility verdict

Each iteration solves two convex QPs. The first realizes the hard tasks as well as the bounds allow,

$$\delta\boldsymbol{q}_1 = \arg\min_{\delta\boldsymbol{q}} \ \tfrac{1}{2}\|J_1\delta\boldsymbol{q} - \boldsymbol{e}_1\|^2 \quad \text{s.t.} \quad \boldsymbol{l} \le \delta\boldsymbol{q} \le \boldsymbol{u}, \ \ G\delta\boldsymbol{q} \le \boldsymbol{g}, \qquad \boldsymbol{r}_1 = J_1\delta\boldsymbol{q}_1 - \boldsymbol{e}_1. \tag{16}$$

The second minimizes the intents without degrading that optimum:

$$\delta\boldsymbol{q}_2 = \arg\min_{\delta\boldsymbol{q}} \ f_2(\delta\boldsymbol{q}) \quad \text{s.t.} \quad U_r^\top J_1 \delta\boldsymbol{q} = U_r^\top J_1 \delta\boldsymbol{q}_1, \ \ \boldsymbol{l} \le \delta\boldsymbol{q} \le \boldsymbol{u}, \ \ G\delta\boldsymbol{q} \le \boldsymbol{g}, \tag{17}$$

where U_r holds the left singular vectors of J_1 with $\sigma_i > \varepsilon\sigma_{\max}$, so that rank-deficient weld stacks (e.g. four latched pads on a closed chain) yield a consistent equality set. The configuration is updated on the manifold, $\boldsymbol{q} \leftarrow \boldsymbol{q} \oplus \delta\boldsymbol{q}_2$ (base rotation by the exponential map, joints and gimbals additively), and the iteration stops when

$$\|\boldsymbol{e}_h^p\| < \varepsilon_p = 1.5 \times 10^{-4}\,\text{m}, \quad \|\boldsymbol{e}_h^r\| < \varepsilon_r = 2 \times 10^{-3}\,\text{rad} \ \ \forall h : \sigma_h = 1, \quad \|\delta\boldsymbol{q}_2\|_\infty < \varepsilon_q = 10^{-4}, \tag{18}$$

or after $K_{\max} = 100$ iterations ($\Delta = 0.12$; the first frames after a release are throttled to $\Delta = 0.003$ and two iterations).

Lexicographic optimality. By construction $\delta \boldsymbol{q}_2$ attains the level-1 residual $\boldsymbol{r}_1$ and, among all steps that do, minimizes f_2. With equality constraints only this reduces to the classical null-space projection $\delta \boldsymbol{q}_2 = \delta \boldsymbol{q}_1 + N \boldsymbol{z}^*$, $N = I - V_r V_r^\top$; the QP form is what admits the box and clearance inequalities at the same priority as the welds.

Feasibility verdict. If at termination $\|\boldsymbol{r}_1\|$ exceeds the tolerances, the frame is reported infeasible together with the active bounds (which joint or gimbal sits at its limit). The solver never resolves such a conflict by moving a latched pad or by overshooting a stop; it surfaces the conflict to the choreographer. This is the mechanism that turns choreography design into a search over a small parameter vector with a binary kinematic verdict per frame (Sec. 5.3).

4.5 Equivalence: bounded gimbal ≡ box-relaxed rotational weld

Let $\boldsymbol{o}_h$ be the gimbal pivot (the centroid of the pad's contact points) and R_h^L the orientation of the limb link above the gimbal. Since the pad is rigidly attached to the wall while latched and the limb is related to it by the gimbal rotation about $\boldsymbol{o}_h$, constraints (13)–(9) are equivalent to

$$\boldsymbol{o}_h(\boldsymbol{q}) = \boldsymbol{o}_h^a, \qquad \left|\log\!\big(R_h^{L\top}(\boldsymbol{q})\, R_h^a\big)_k\right| \le \gamma_{\max}, \quad k = 1,2,3, \tag{19}$$

i.e. a 3-DoF positional weld of the limb at the pivot plus a componentwise box on its orientation error. The bounded-variable form is preferred in the implementation because it reuses the plant's own Jacobians and pivot geometry without a small-angle approximation, and because the recorded $\boldsymbol{\gamma}_h(t)$ directly predicts how much of the physical travel the reference consumes.

4.6 Weighted single-level variant (baseline)

The baseline replaces the hierarchy by one weighted damped least-squares step,

$$\delta \boldsymbol{q} = \Big(\sum_i J_i^\top W_i^2 J_i + D + \lambda^2 I\Big)^{-1} \Big(\sum_i J_i^\top W_i^2 \boldsymbol{e}_i + \boldsymbol{d}\Big), \tag{20}$$

where the weld rows enter with $W = \mathrm{diag}(1000, 1000, 1000, 500, 500, 500)$, the gimbals are locked at zero, and $(D, \boldsymbol{d})$ collect the posture and limit-band springs; joint limits are enforced by clipping after the step. Because weights trade off, the weld leaks: the latched pad drifts by an amount that grows with the intent weights (measured up to 3.4 mm inside the swings of the shipped cycle and 1.6 mm on the mount prologue under a 2× weld weight; re-anchoring at every segment boundary hides the accumulated drift from the dwells), and an over-constrained arm resolves a rotational demand by translating the pad off its anchor rather than by reporting infeasibility. Section 7 compares the two.

4.7 Cycle closure, tiling and velocities

The last swing returns the robot to the stance; residual solver error is removed by a closure blend over the final $n_c = \lceil 0.4 N_{\text{last}} \rceil$ frames,

$$\boldsymbol{\Delta} = (\boldsymbol{q}^0 \oplus h\hat{\boldsymbol{z}}) \ominus \boldsymbol{q}_T, \qquad \boldsymbol{q}_{T-n_c+i} \leftarrow \boldsymbol{q}_{T-n_c+i} \oplus S\Big(\tfrac{i+1}{n_c}\Big)\, \boldsymbol{\Delta}, \quad i = 0, \dots, n_c - 1, \tag{21}$$

so that $\boldsymbol{q}(T) = \boldsymbol{q}^0 \oplus h\hat{\boldsymbol{z}}$ exactly: same joint angles, base translated by one step. Cycles tile as $\boldsymbol{q}^{(k)} = \boldsymbol{q} \oplus kh\hat{\boldsymbol{z}}$ with anchors shifted alike. Velocities are forward differences on the loop, $\dot{\boldsymbol{\theta}}_t = f_s(\boldsymbol{\theta}_{t+1} - \boldsymbol{\theta}_t)$ and $\boldsymbol{\omega}_t = f_s \log(q_{t+1} \otimes q_t^{-1})$, where frame T is frame 0 shifted, so the wrap velocity is the true loop velocity.

5 Physics gates

The generator is purely kinematic. Two audits certify that the result is physically holdable; both are cheap, deterministic, and independent of any learned policy and of the simulator's constraint stiffness.

5.1 Static force-existence audit on the limit surface

At each frame the robot is posed at $\boldsymbol{q}(t)$ (gimbals at their recorded angles) and we ask whether *some* allocation of contact forces balances gravity within the pad envelope and the actuator limits. Unknowns are the pad-on-wall forces $\boldsymbol{F}_h$ of latched pads, unilateral wall presses $\boldsymbol{C}_j$ (raking pad faces, braced toes), floor forces $\boldsymbol{P}_k$ (mount prologues only) and a slack s:

$$\begin{aligned} \min_{\{\boldsymbol{F}_h\},\{\boldsymbol{C}_j\},\{\boldsymbol{P}_k\},s} \quad & s + \epsilon\|\boldsymbol{z}\|^2 \\ \text{s.t.} \quad & \textstyle\sum_h \boldsymbol{F}_h + \sum_j \boldsymbol{C}_j + \sum_k \boldsymbol{P}_k = m\,\boldsymbol{g}, \\ & \textstyle\sum_h (\boldsymbol{r}_h - \boldsymbol{r}_{\mathrm{com}}) \times \boldsymbol{F}_h + \sum_j (\boldsymbol{r}_j - \boldsymbol{r}_{\mathrm{com}}) \times \boldsymbol{C}_j + \sum_k (\boldsymbol{r}_k - \boldsymbol{r}_{\mathrm{com}}) \times \boldsymbol{P}_k = \boldsymbol{0}, \\ & u_h(\boldsymbol{F}_h) \le s \quad \forall h : \sigma_h = 1, \\ & \left|\boldsymbol{\tau}_0 + \textstyle\sum_h J_h^\top \boldsymbol{F}_h + \sum_j J_j^\top \boldsymbol{C}_j + \sum_k J_k^\top \boldsymbol{P}_k\right| \le \boldsymbol{\tau}_{\max}, \\ & c_{j,n} \ge 0,\ \|\boldsymbol{c}_{j,t}\| \le \mu_w c_{j,n}; \qquad P_{k,z} \le 0,\ \|\boldsymbol{P}_{k,xy}\| \le \mu_g(-P_{k,z}), \end{aligned} \tag{22}$$

where $\boldsymbol{\tau}_0$ is the gravity torque at zero velocity minus passive joint torques, $\boldsymbol{\tau}_{\max}$ the deployment torque limits, and the pad utilization is

$$u_h(\boldsymbol{F}) = \frac{\sqrt{f_t^2 + f_l^2 + \max(f_n, 0)^2}}{F_0\ \max\big(R(\vartheta, \varphi),\, r_{\mathrm{floor}}\big)}, \qquad \vartheta = \operatorname{atan2}(|f_l|, f_t), \quad \varphi = \operatorname{atan2}\big(\max(f_n, 0), \sqrt{f_t^2 + f_l^2}\big), \tag{23}$$

with $R(\vartheta, \varphi) \in [0, 1]$ the normalized Wang limit surface (tabulated, bilinearly interpolated), F_0 the pure-rake capacity (400 N, the minimum of the training-time randomization range) and compression $f_n < 0$ carried by ordinary contact. Local pad moments are zero by construction (point forces at the pad centers), so a passing frame cannot be relying on the weld as a flange. The audit *passes* iff $s^\star \le 1$ at every frame; the shipped cycle reports $\max_t s^\star = 0.877$. Margins are built in: $\mu_w = 0.5$ (simulated 0.8), $\mu_g = 0.6$ (simulated 1.0).

5.2 Schedule-timing audit

Let $\tau_{\mathrm{cd}} = 0.20\,\mathrm{s}$ be the re-engage cooldown, $\tau_{\mathrm{db}} = 0.06\,\mathrm{s}$ the press debounce and $\tau_{\mathrm{mv}} = 0.25\,\mathrm{s}$ a floor for the actual pad travel. Over the circular schedule, every OFF window W of every pad must satisfy

$$|W| \ge \tau_{\mathrm{cd}} + \tau_{\mathrm{db}} + \tau_{\mathrm{mv}}, \tag{24}$$

and for every regrip of pad h at t_r and the next release t_f of any other pad,

$$t_f - t_r \ge \tau_{\mathrm{ramp}} = 0.25\,\mathrm{s} \qquad (\text{warning if } t_f - t_r < \tau_{\mathrm{ramp}} + 0.3\,|W|), \tag{25}$$

because a freshly latched pad holds only a fraction of its capacity until it has recruited under load. A cycle that violated this by 0.05 s (`fast` timing, 0.2 s seat holds) trained policies that abandoned the immature foot; restoring the dwells (Table 1) fixed it without any change of pose.

5.3 Design loop

The choreography is a vector $\boldsymbol{\psi}$ of about fifteen scalars (step height, bulges, press and lateral shifts, rake length, brace timing) plus the segment durations. Generation with the IK verdict, the static

audit and the timing audit takes minutes and is bitwise reproducible, so $\boldsymbol{\psi}$ is tuned in a tight loop and the final recipe is stored with the clip. Because the generator does not read any runtime physics parameter, recalibrating the simulator's pad model does not invalidate the reference: after the 4° suspension recalibration the regenerated clip was identical to the previous one.

6 Consuming the reference in reinforcement learning

The clip is resampled to the policy step Δt and indexed by a phase $k \in \{0, \dots, C-1\}$ that advances every step and wraps with the base offset increased by h. Episodes start by reference-state initialization: a uniformly sampled phase, the robot placed at $\boldsymbol{q}_k$ with joint noise and a height jitter along the wall, and the pad latches seeded from $\boldsymbol{\sigma}_k$. For a tilted wall the reference is rotated about the wall base line; joint angles are frame-invariant.

Imitation rewards use the exponential kernel $r = \exp(-e/\sigma^2)$ on the mean squared joint-angle and joint-velocity errors, on the root-frame end-effector positions, and on the root-orientation geodesic error; each pose reward is multiplied by a schedule gate $g = g_0 + (1 - g_0)\big(1 - \overline{\text{mismatch}}\big)$, where a pad's mismatch $|\sigma_h - \hat{\sigma}_h|$ is discounted linearly within a grace fraction of its scheduled window around every edge. Schedule mismatch is also billed directly, landings are rewarded once per window by their distance to $\boldsymbol{a}_h$, and climb progress is rewarded relative to the clip's own rate hf_s/C. The base's world position is never tracked: the reference dictates *shape and schedule*, the policy learns force allocation and balance under domain-randomized pad capacities.

7 Properties and ablations

The construction guarantees, per frame and by construction rather than by tuning: (i) latched pads are stationary within $\varepsilon_p, \varepsilon_r$; (ii) joints and gimbals stay strictly inside their limits; (iii) released pads and toes do not penetrate the wall; (iv) a moment-free static force allocation inside the limit surface and torque limits exists; (v) dwell times cover the engage pipeline and the recruitment ramp; (vi) the cycle closes exactly. We report the following ablations [TBD]:

- **Weighted vs. lexicographic IK** (Sec. 4.6 vs. 4): maximum latched-pad drift, worst joint-limit margin, number of frames reported infeasible, and the reachable envelope of the choreography (highest feasible hand anchor at fixed standoff).
- **Locked vs. bounded gimbal** ($\gamma_{\max} \in \{0, 1, 2, 2.5\}^\circ$): pad drift and infeasible frames on the mount prologue, where the closed chain of two welded 4-DoF arms is over-constrained.
- **Physics gates on/off**: policies trained on the `fast` (timing-violating) and `paced` cycles; retargeted style clips as a further baseline.
- **Downstream**: engage match, per-pad occupancy, climb rate and fall rate of the trained policy on the vertical wall.